17. ORGANISATION OF METABOLIC FIELDS

DURATION : 08:21

STUDY SHEET

COURSE SUMMARY

In this video, we explore the organisation of metabolic fields during embryonic development, starting from the embryonic knob and the formation of the blastocoel. Key concepts include the introduction of permeation, permeation, and infusion fields, which determine how trophic information is distributed and influences tissue transformation. The video also details the role of the embryonic pedicle, which guides the polarised trophic supply, as well as the dynamics that lead to the formation of the 'S' shape of the embryonic tissue. This teaching is essential for understanding the foundations of biodynamic embryology and its application in osteopathy.

ORGANISATION OF METABOLIC FIELDS

During **embryonic development**, we start from an embryonic knob and a cavity, the **blastocoel**.

At the time of **implantation** in the uterine lining, a second cavity appears. At this stage, two types of cells exchange: one faces the older cavity, and it is at this point that Blech-Schmidt introduces the concepts of **endocyst**, **epiblast**, and **hypoblast**.

When the third chamber appears, a new system is established, characterised by a **differential growth rate** between the outside and the inside. Inside, growth is slightly slower, as the **trophic supply** arrives less quickly.

An important phenomenon occurs at the tissue level, inducing new information. These are what are called **metabolic fields** of **precession**, **permeation**, **parmeation**, and **infusion** in the trophic supply of information. In other words, it is an organisation of how trophic information, the **primitive nourishment**, distributes itself in the tissues to transform them.

Blech-Schmidt understood how this happened and introduced new notions that are only found in **biodynamic embryology**. Information is distributed both by what is called a **parmeation field**, a **permeation field**, or an **infusion field**.

METABOLIC FIELDS: DEFINITIONS AND DIFFUSION MECHANISMS

What do these terms mean?

Let's take a layer of cells. Trophic information now arrives in a much more specific and polarised way. With the appearance of the **vitelline cavity**, the **amniotic cavity**, and the **extraembryonic coelom**, what is called an **embryonic pedicle** is organised. This means that trophic information no longer comes from everywhere, but in an oriented way.

Information no longer arrives from all sides, but from one side, called the **embryonic pedicle**. If we observe a group of cells receiving the trophic supply, we see that this happens in three different ways from the embryonic pedicle:

1. **Parmeation field**: Information can pass along the membranes, in a parallel and linear fashion.

2. **Permeation field**: Information can pass between the cells.

3. **Infusion field**: The cell is there, and it wants to "eat" directly.

This introduces a very important movement that allows the transition from the **zygote** to a **primitive foetus**. Initially, trophic information is general, without any real connection. But when the passage occurs from the blastocoel to a vitelline cavity and an extraembryonic coelom, a concentration occurs. There is no longer just a link between this small mass and the outside part, later forming the **placenta**. This is called the **embryonic pedicle**, or the **primitive umbilical cord**, a trophic site.

ORGANISATION OF THE EMBRYO AND METABOLIC MOVEMENT

Now let's take an embryo at this stage. We have a layer of cells forming the **amniotic cavity** and a layer of cells below, the **vitelline cavity**. The whole is connected to an embryonic pedicle and is located in a cavity, the **extraembryonic coelom**.

At that moment, a trophic source is organised from the outside inwards, but in a polarised way. The upper part shows greater metabolic activity, due to its anterior position. Information is concentrated in a parmeation field.

At the same time, there is a central metabolic field of parmeation, infusion, and permeation, organising the metabolic movement of the embryo. The **vitelline cavity** moves backwards and the **amniotic cavity** forwards, introducing an **S-shape** within the germinal disc.

FORMATION OF THE "S" SHAPE AND REPRESENTATION ERROR

This movement transforms the tissue, which was initially flat, into a tissue with an S-shape. Metabolic activity brings the amniotic cavity forwards and the vitelline cavity backwards.

It is crucial to note the error between the representations of these two drawings. The extraembryonic coelom must be much larger. The error lies in the relative size of the structures. It is necessary to resize the drawing to have a more accurate view, because it is in this passage between the extraembryonic coelom and the vitelline cavity that the pedicle is formed.

In this pedicle, a movement of **permeation**, **parmeation**, and **infusion** develops, creating a forward movement of the amniotic cavity, slightly less pronounced below. This leads to a tilt, and this tilting movement creates an S-shape within the epiblast and endoblast.

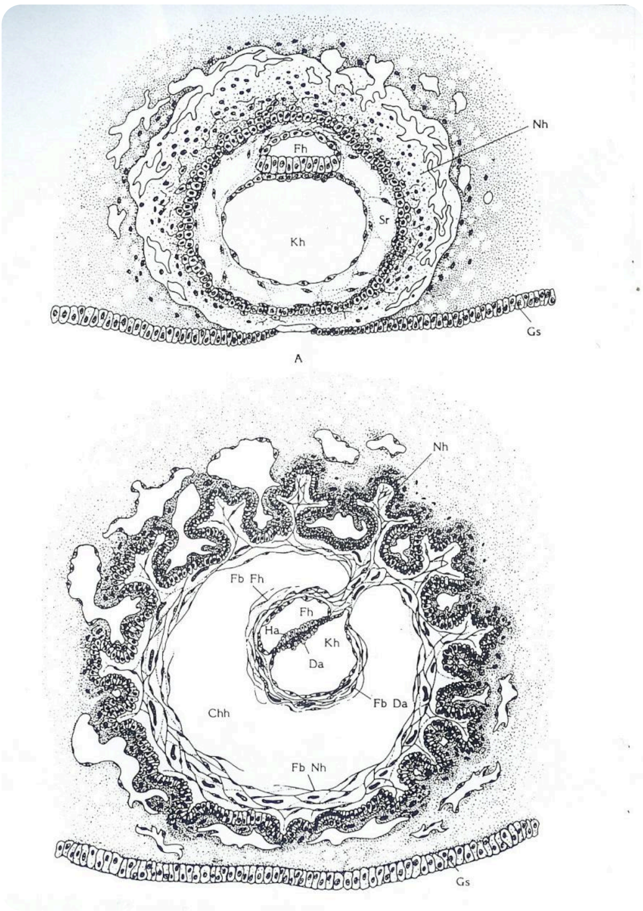

FIG. — Calcium salts